DNS Resolver – Phase 3 Extension: CNAME Resolution

Computer Networks Course Project · Phase 3 Implementation

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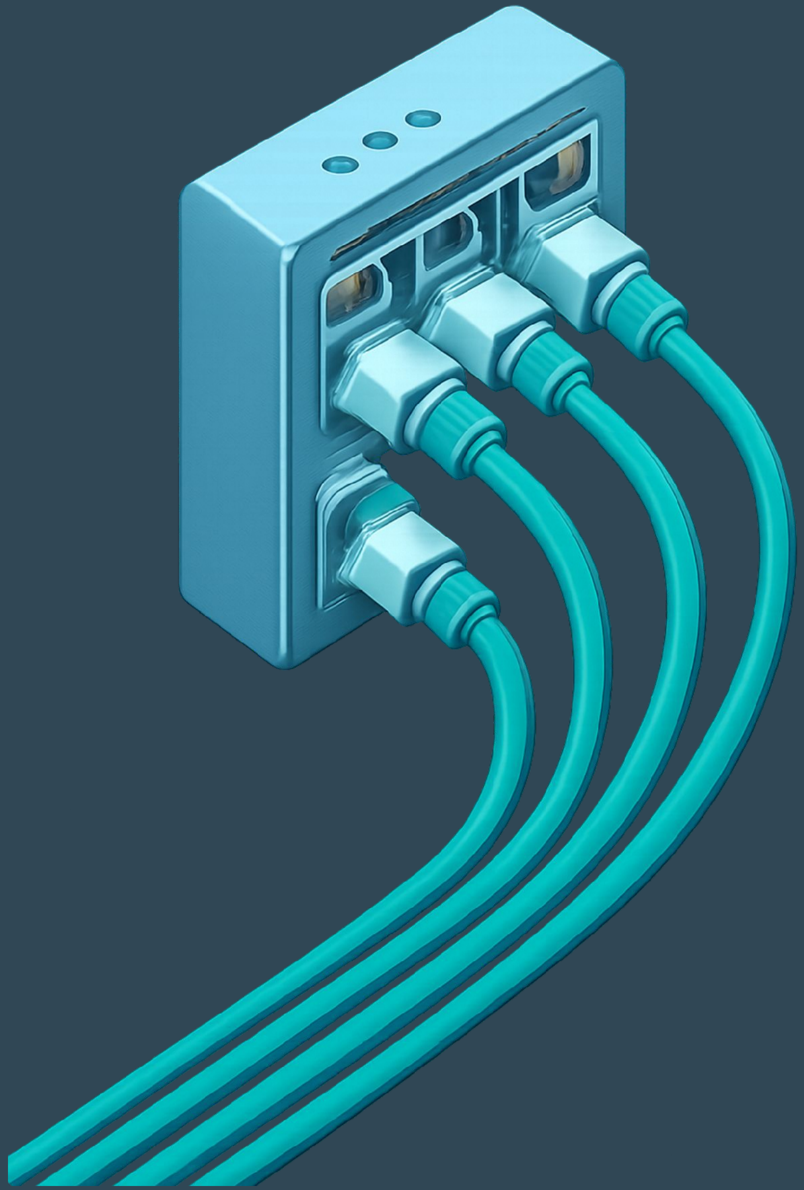
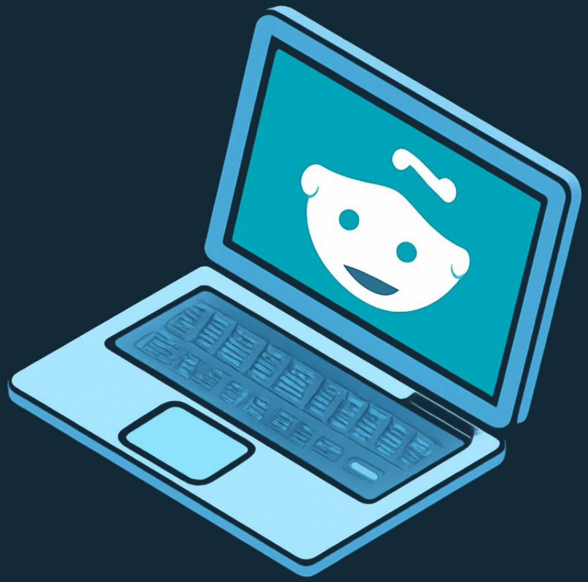


The Problem: CNAME Records

- Phase 1–2 only handled **A records** — direct domain-to-IP mappings.
- Real DNS responses often return **CNAME records**, pointing to another domain.
- Without CNAME support, resolution fails.



REAL-WORLD EXAMPLE

www.reddit.com

`www.reddit.com` returns `CNAME → reddit.map.fastly.net` → **151.101.41.140**

Resolution Flow with CNAME

The resolver must follow the alias chain until it reaches an A record. Each hop traverses the full DNS hierarchy.

1. Query `www.reddit.com` at root, then `.com`, then the authoritative server.
2. Receive a `CNAME` to `reddit.map.fastly.net`.
3. Repeat the lookup for `reddit.map.fastly.net` through the DNS hierarchy.
4. Return the final A record: `151.101.41.140`.

Each alias requires a full recursive resolution pass — the resolver must be prepared to restart the process from the root servers with the new canonical name.

Implementation: Four Code Changes

- Define `TYPE_CNAME = 5` to identify CNAME records in DNS responses.
- Update the parser to decode CNAME record data as compressed DNS names.
- Add `get_cname(packet)` to search the answer section and return the canonical name.
- Resolve CNAMEs recursively by calling `resolve()` with the canonical name.

KEY CODE SNIPPET

Following the Alias

When a CNAME appears, the resolver logs it and immediately follows the alias with a recursive lookup.

```
elif cname := get_cname(response):
    print(f"Following CNAME: {domain_name} →
{cname}")
    return resolve(cname, TYPE_A)
```

The code detects a CNAME response, prints the hop, and recursively resolves the canonical name.

Demo: Resolver in Action

```
Last login: Sun Aug  2 12:39:34 on ttys002
amiiir@Amirs-MacBook-Pro ~ % cd ~/dns-resolver
amiiir@Amirs-MacBook-Pro dns-resolver % python3 resolver.py www.reddit.com
Querying 198.41.0.4 for www.reddit.com
Querying 192.41.162.30 for www.reddit.com
Querying 205.251.193.122 for www.reddit.com
Following CNAME: www.reddit.com -> reddit.map.fastly.net
Querying 198.41.0.4 for reddit.map.fastly.net
Querying 192.55.83.30 for reddit.map.fastly.net
Querying 23.235.32.32 for reddit.map.fastly.net
151.101.41.140
amiiir@Amirs-MacBook-Pro dns-resolver % dig www.reddit.com CNAME

; <<>> DiG 9.10.6 <<>> www.reddit.com CNAME
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 42763
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;www.reddit.com.                        IN      CNAME

;; ANSWER SECTION:
www.reddit.com.         176     IN      CNAME   reddit.map.fastly.net.

;; Query time: 15 msec
;; SERVER: 10.0.0.1#53(10.0.0.1)
;; WHEN: Mon Aug 03 00:24:27 PDT 2026
;; MSG SIZE  rcvd: 78

amiiir@Amirs-MacBook-Pro dns-resolver %
```

Verification: Consistency with `dig`

To validate correctness, the resolver output was compared against the industry-standard `dig` tool:

```
$ dig www.reddit.com CNAME

;; ANSWER SECTION:
www.reddit.com.  300  IN  CNAME
reddit.map.fastly.net.
```

It matches `dig`'s CNAME result.

What This Confirms

- Correct parsing of CNAME record type (type code 5)
- Accurate DNS name decompression from the wire format
- Resolver follows the same alias chain as production tools
- Output matches expected real-world DNS behavior

STANDARDS COMPLIANCE

RFC 1034 Compliance

Phase 3 is grounded in the foundational DNS standard. **RFC 1034, Section 3.6.2** defines the Canonical Name (CNAME) resource record and its required behavior:

> The resolver must continue resolving the canonical name until reaching the requested resource record type.

Phase 3 implements this specification faithfully.